

BUS123 MATH M04 · Markups, Markdowns, and Break-Even

Connect to Prior Knowledge

In the discounts lesson, you started with a list price and reduced it. In this lesson, you often start with cost and build up to a selling price. The percentage mechanics are familiar, but the business question changes: will this price cover the cost, reflect the risk, and still make sense to the customer?

Why Markup Matters

Markup is the amount added to cost to create a selling price. Anchor & Oak Events may know the ingredient and packaging cost for a tasting box, but that cost is not the price. The price must also cover labor, event preparation, waste, and profit. A 45% markup on cost means the selling price equals cost plus 45% of cost. In a spreadsheet, this is usually modeled by placing cost in one cell, markup rate in another, and writing a formula that multiplies cost by one plus the markup rate.

Markup on Cost vs. Markup on Selling Price

A common source of confusion is the base. Markup on cost uses cost as the denominator. Markup on selling price uses the selling price as the denominator. A product that costs \$18 and sells for \$26.10 has \$8.10 of markup. That is 45% of cost, but about 31% of selling price. Same dollars, different percentage. Whenever you read a business problem, pause and identify the base before choosing a formula.

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Markdowns and Perishables

Markdowns reduce price after the original price has been set. For an event business, markdowns can help fill late seats or sell perishable items before they lose value. A markdown is not automatically a failure. It is a tradeoff: lower margin per sale in exchange for more cash recovery, fuller attendance, or less waste.

Waste Changes the Pricing Problem

Perishable products require extra care because not every produced unit may become a sellable unit. If Anchor & Oak produces 120 tasting boxes and expects 15% waste, only 102 boxes are expected to be sold. The cost of all 120 boxes still needs to be recovered from the 102 sellable boxes. This is why a price that looks profitable before waste can become too low after waste is considered.

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Break-Even Logic

Break-even is the point where total revenue equals total cost. Before break-even, the business is losing money on the event. After break-even, each additional unit contributes toward profit. The key idea is contribution margin: selling price minus variable cost per unit. Contribution margin is not the same as profit, because fixed costs still need to be covered first.

Anchor & Oak Example

Suppose an event has \$3,600 in fixed cost, a \$48 ticket price, and \$27 variable cost per guest. Each guest contributes \$21 toward fixed cost and profit. The break-even guest count is \$3,600 divided by \$21, rounded up to 172 guests. The rounding matters: selling to 171 guests would still leave a small loss, while 172 guests crosses the break-even point.

Worked Example 1: Markup With Waste

Anchor & Oak plans to produce 120 tasting boxes. Each box costs \$18 to prepare, and the target markup is 45% on cost. If every box could be sold, the target revenue would be $120 \times \$18 \times 1.45$, or \$3,132. But the business expects 15% waste, so only 102 boxes are likely to be sold. The required price per sellable box is $\$3,132 \div 102$, or \$30.71. Notice what happened: the price rose because fewer sellable boxes must recover the same production cost.

Excel pattern: `=ProducedUnits*CostPerUnit*(1+MarkupRate)` and `=TargetRevenue/SellableUnits`

Worked Example 2: Break-Even

Anchor & Oak is considering a tasting event with \$3,600 of fixed cost, a \$48 ticket price, and \$27 variable cost per guest. The contribution margin per guest is \$21. Break-even guests equals fixed cost divided by contribution margin: $\$3,600 \div \$21 = 171.43$. Because the business cannot sell a fraction of a guest ticket, Excel should round up to 172 guests.

Excel pattern: `=ROUNDUP(FixedCost/(TicketPrice-VariableCost),0)`

BUS123 MATH M04 · Formula Reference and Practice

Formula Reference

Concept	Formula
Markup amount	Selling price - Cost
Markup on cost	Markup amount / Cost
Selling price from markup on cost	Cost * (1 + Markup rate)
Markdown price	Original price * (1 - Markdown rate)
Sellable units	Produced units * (1 - Waste rate)
Contribution margin	Selling price - Variable cost per unit
Break-even units	Fixed costs / Contribution margin per unit
Break-even revenue	Break-even units * Selling price

Check Your Understanding

A dessert box costs \$20 and is marked up 40% on cost. What is the selling price?

A \$50 ticket is marked down 20%. What is the new price?

Anchor & Oak produces 160 favors and expects 10% waste. How many are sellable?

An event has a \$35 ticket price and \$18 variable cost per guest. What is contribution margin?

Fixed cost is \$2,400 and contribution margin is \$16. How many units are needed to break even?

BUS123 MATH M04 · Vocabulary

Key Vocabulary

Markup	Amount added to cost to create a selling price.
Markup on cost	Markup measured as a percentage of cost.
Markdown	Reduction from an original selling price.
Perishable inventory	Goods or capacity that lose value over time.
Variable cost	Cost that changes with each unit or guest.
Fixed cost	Cost that does not change with each unit in the short run.
Contribution margin	Selling price minus variable cost per unit.
Break-even point	Sales volume where revenue equals total cost.